034IN -FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL

Second Partial Examination - A.Y. 2024/2025

June 6, 2025

Design of a Strictly Proper Controller for a Plant with a Parameter Affected by Interval Uncertainty

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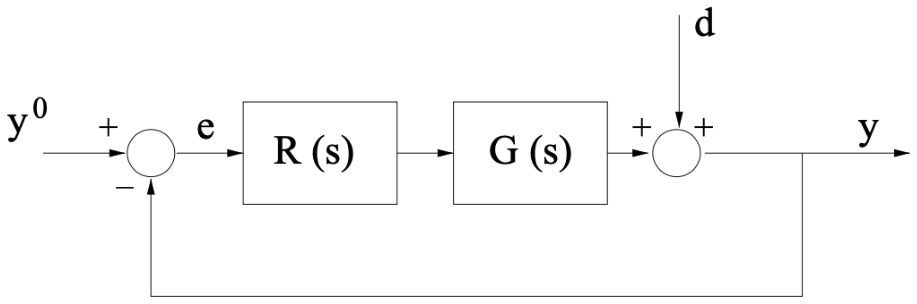
Description of the Feedback Control System

Consider the feedback control system described by the block-scheme shown in the following figure



where $G(s)$ is given by

$$G(s) = 20\,\frac{(1+5\,s)\left(1+\frac{s}{250}\right)}{\left(1+\frac{1.6}{\Omega}s+\frac{s^2}{\Omega^2}\right)\left(1+\frac{s}{100}\right)}\,, \qquad \Omega \in [9\,,\,16]\ \text{rad/sec}$$

Note: This means that the system has a transfer function with two complex conjugate poles, whose damping is known exactly ($\xi = 0.8$) , but whose natural pulsation $\omega_n = \Omega$ is unknown. All that is known is that the natural pulsation ω_n of the complex conjugate poles takes on a value contained within the assigned range $[9\,,\,16]$ rad/s. The zeros and the remaining pole are perfectly known.

Question

Q1: Design a **strictly proper controller** $C(s)$ such that all following requirements are simultaneously met:

- **Zero steady-state error** in unit step response (closed-loop) $e(\infty) = 0$, against disturbances given by

$$\begin{cases} y^o(t) &= 1(t) \\ d(t) &= A \cdot 1(t)\,,\ A \in \mathbb{R} \end{cases} \quad \Longrightarrow \quad e_\infty = 0$$

where $\mathbf{1}(t)$ denotes the unit step function;

- **phase margin** greater than or equal to 60°: $\varphi_m \geq 60^\circ$;
- **crossover angular frequency** ω_c between 1 rad/sec and 5 rad/sec: $1 \leq \omega_c \leq 5$

Hint

How to handle the unknown natural pulsation $\omega_n = \Omega$? A good practice is to identify the **worst-case scenario** and design the controller $C(s)$ according to that scenario.

Solution

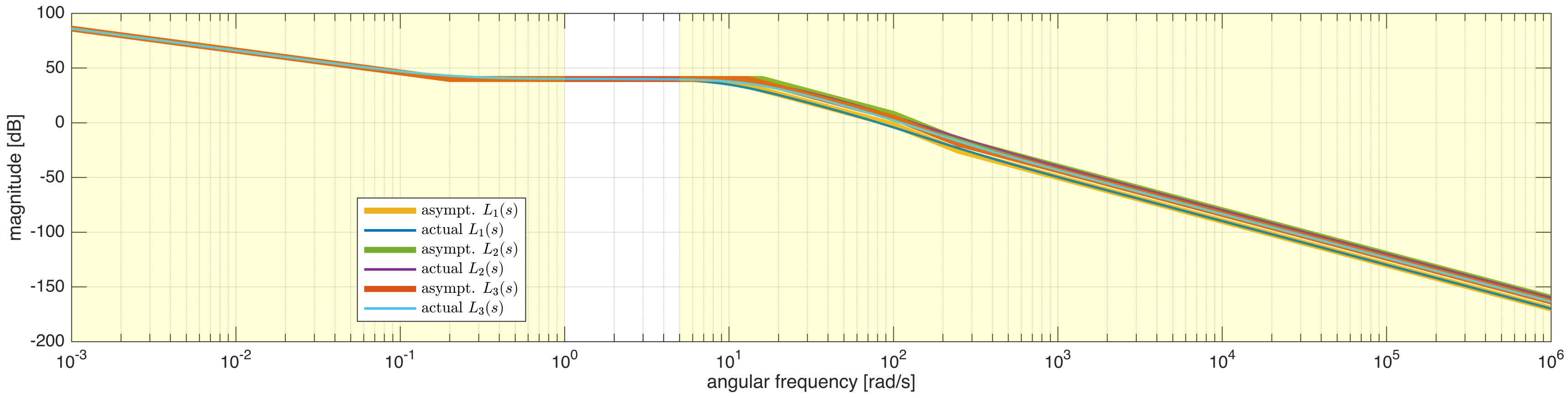
Initialisation code (clearing the MATLAB workspace, adding the path of the BodeDiagram subfolder to the MATLAB search-path)

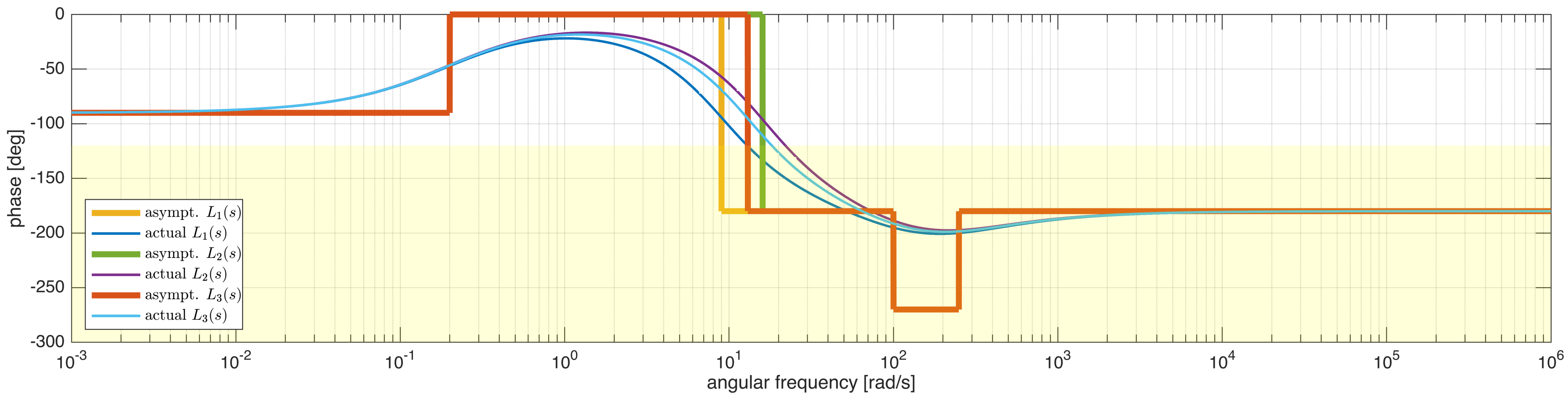
```
clear variables

% adding folders (and subfolders) to search path
addpath(genpath('BodeDiagram/'))

% let's define the transfer function builder element
s=tf('s');
```

Preliminary Analysis

```
omMIN = 9; omMAX = 16;
omR = 13;

% static regulator
muC = 1;
C1s = muC/s;

Gs1 = 20*(1+5*s)*(1+s/250)/(1+(1.6/omMIN)*s +s^2/omMIN^2)/(1+s/100);
Gs2 = 20*(1+5*s)*(1+s/250)/(1+(1.6/omMAX)*s +s^2/omMAX^2)/(1+s/100);
Gs3 = 20*(1+5*s)*(1+s/250)/(1+(1.6/omR)*s +s^2/omR^2)/(1+s/100);

Ls1 = C1s*Gs1;
Ls2 = C1s*Gs2;
Ls3 = C1s*Gs3;

% the Bode diagrams

Bcolors = [0 0.4470 0.7410; ...
           0.9290 0.6940 0.1250; ...
           0.4940 0.1840 0.5560; ...
           0.4660 0.6740 0.1880; ...
           0.3010 0.7450 0.9330; ...
           0.8500 0.3250 0.0980; ...
           0.6350 0.0780 0.1840]; % some different colors


hf = figure('Units','normalized','Position',[0.1, 0.1, 0.95, 0.85]);
drawBodediagrams(Ls1, [], Bcolors(2,:) , 3.5, [], Bcolors(1,:) ,1.5, [], hf);
drawBodediagrams(Ls2, [], Bcolors(4,:) , 3.5, [], Bcolors(3,:) ,1.5, [], hf);
[ha1, ha2] = drawBodediagrams(Ls3, [], Bcolors(6,:) , 3.5, [], Bcolors(5,:) ,1.5, [], hf);

phaseM = +60; % the desired minimum phase margin
feas_Ls_Phase = -180+phaseM; % the minimum value of the arg L(j omega)
                             % to guarantee the feasibility of the design

YLIMS = ylim(ha2);
OMLIMs = xlim(ha2);

% the (Q1.2) constraint
minOmega1 = OMLIMs(1);
maxOmega1 = OMLIMs(2);
PhLIM = feas_Ls_Phase;

Vert1 = [minOmega1 ,  YLIMS(1); minOmega1 ,PhLIM; ...
         maxOmega1 , PhLIM; maxOmega1 , YLIMS(1)];
Faces1 = [1  2  3  4];
hold on;
patch(ha2, 'Faces',Faces1,'Vertices',Vert1,...
        'FaceColor','yellow','FaceAlpha',.15, ...
        'EdgeColor', 'none'); % the pale yellow region is the forbidden one

omC1 = 1; omC2 = 5; % constraint (Q1.3)
YLIMS = ylim(ha1);
OMLIMs = xlim(ha1);
minOmega1 = OMLIMs(1);
maxOmega1 = OMLIMs(2);
Vert2 = [minOmega1 ,  YLIMS(1); omC1 ,YLIMS(1); ...
         omC1 , YLIMS(2); minOmega1 , YLIMS(2)];
Faces2 = [1  2  3  4];
hold on;
patch(ha1, 'Faces',Faces2,'Vertices',Vert2,...
```

```
        'FaceColor','yellow','FaceAlpha',.15, ...
        'EdgeColor', 'none'); % the pale yellow region is the forbidden one

Vert3 = [omC2 ,  YLIMS(1); maxOmega1 ,YLIMS(1); ...
         maxOmega1 , YLIMS(2); omC2 , YLIMS(2)];
Faces3 = [1  2  3  4];
hold on;
patch(ha1, 'Faces',Faces3,'Vertices',Vert3,...
        'FaceColor','yellow','FaceAlpha',.15, ...
        'EdgeColor', 'none'); % the pale yellow region is the forbidden one


legend(ha1, 'asympt. $L_1(s)$', 'actual $L_1(s)$', ...
            'asympt. $L_2(s)$', 'actual $L_2(s)$', ...
            'asympt. $L_3(s)$', 'actual $L_3(s)$', ...
            'Interpreter', 'latex', 'Location', 'best')
legend(ha2, 'asympt. $L_1(s)$', 'actual $L_1(s)$', ...
            'asympt. $L_2(s)$', 'actual $L_2(s)$', ...
            'asympt. $L_3(s)$', 'actual $L_3(s)$', ...
            'Interpreter', 'latex', 'Location', 'southwest')
```






```
% the design problem is feasible!
% the worst-case scenario corresponds to Ls1 --> the phase is the lowest
% one
```

Tuning the Controller

According to (Q1.3) and (Q1.2), any angular frequency $\omega \in \left[1\ ,\ 5\right]$ rad/sec allows to fulfil both the requirements.

Let's tune the gain constant μ_C with the aim to obtain $\bar{\omega} = 2$ rad/sec as crossover frequency in the worst-case scenario

$$\begin{cases} C_1(s) &= \dfrac{\mu_C}{s} \\ G_1(s) &= 20\,\dfrac{(1+5\,s)\left(1+\dfrac{s}{250}\right)}{\left(1+\dfrac{1.6}{\Omega}s+\dfrac{s^2}{\Omega^2}\right)\left(1+\dfrac{s}{100}\right)}, \qquad \Omega = 9 \text{ rad/sec} \end{cases} \implies L_1(s) = 20\,\mu_C\,\frac{(1+5\,s)\left(1+\dfrac{s}{250}\right)}{s\left(1+\dfrac{16}{90}s+\dfrac{s^2}{81}\right)\left(1+\dfrac{s}{100}\right)}$$

Thus

$$\omega_c = 2 \implies |L_1(j2)| = 1 = 20\,\mu_C\left|\frac{(1+j10)\left(1+\dfrac{j2}{250}\right)}{j2\left(1+\dfrac{16}{90}j2-\dfrac{4}{81}\right)\left(1+\dfrac{j2}{100}\right)}\right| \implies \mu_C = ?$$

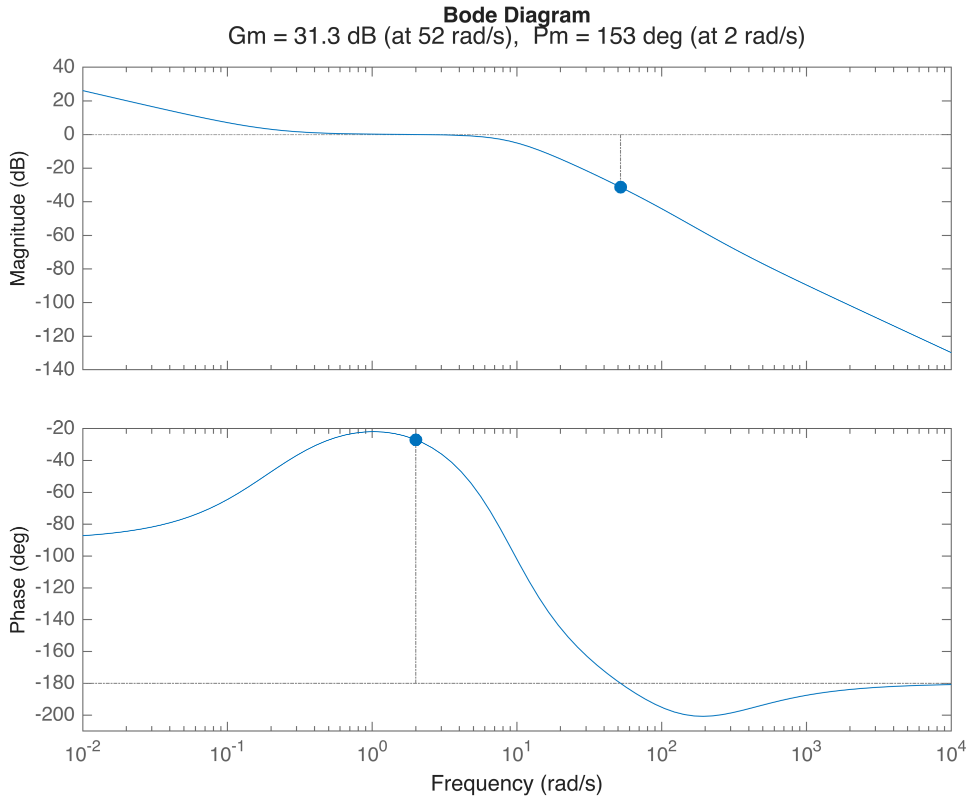
```
barOMc1 = 1j*2;
mag_L1s_barOMc1 = abs(freqresp(Ls1, barOMc1));
```

```
muC_VAL = 1/mag_L1s_barOMc1
```

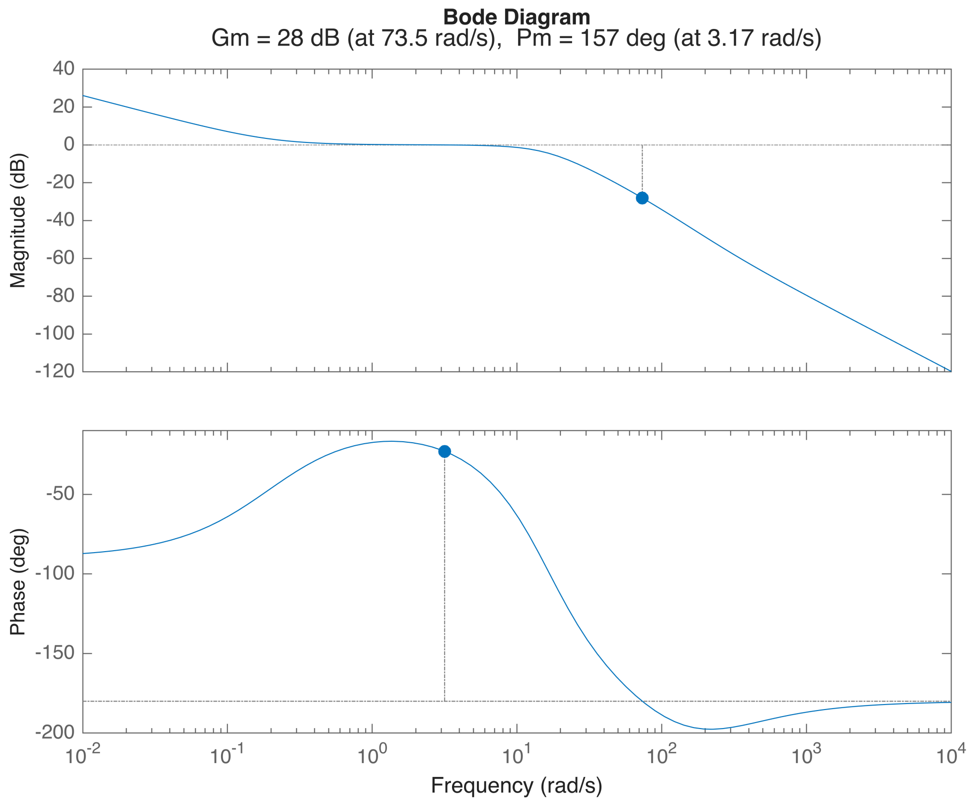
```
muC_VAL =
0.0101
```

Verify the performance of the controller $C_1(s)$ in every scenario

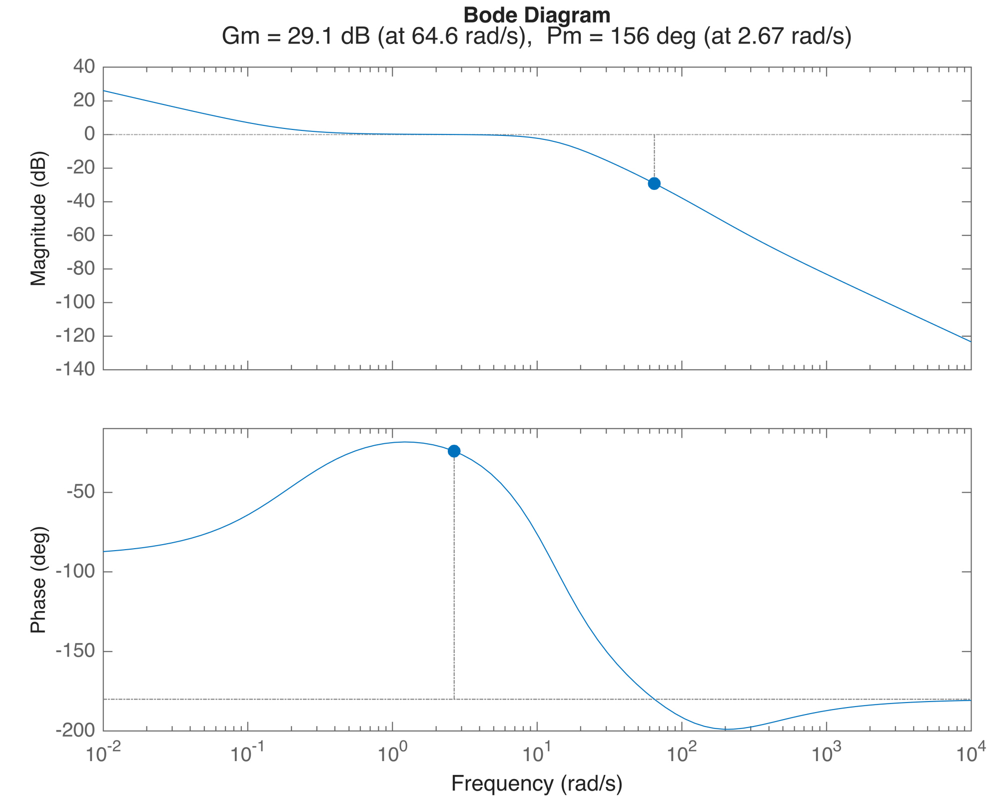
```
C1s = muC_VAL/s;

Ls1 = C1s*Gs1;
Ls2 = C1s*Gs2;
Ls3 = C1s*Gs3;

figure;margin(Ls1)
```




```
figure; margin(Ls2)
```



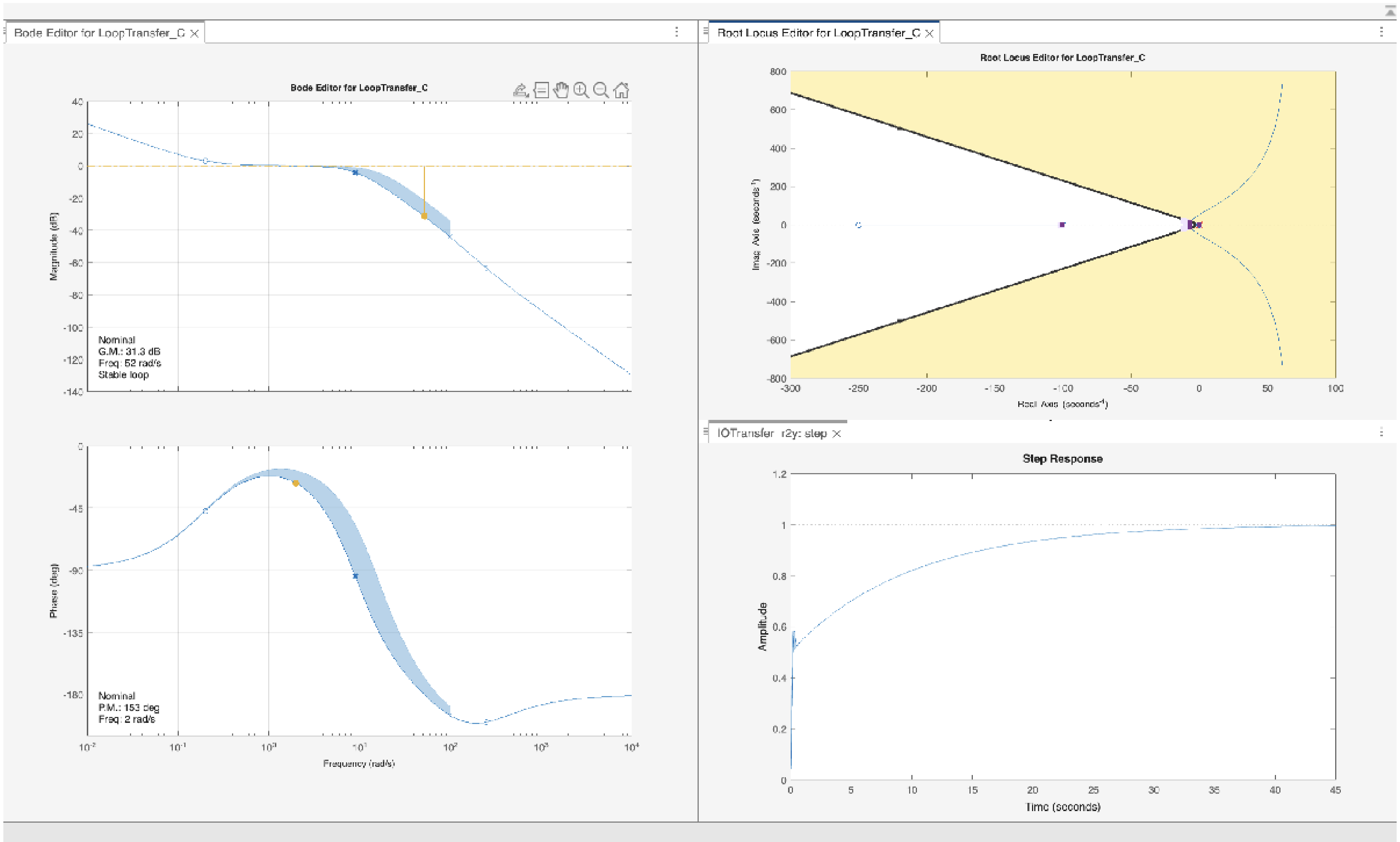

```
figure;margin(Ls3)
```



In conclusion, the controller $C_1(s)$ allows to fulfil every requirement.

Tuning the Controller using the controlSystemDesigner App

```
% let's create a multi-model structure
Gs = stack(1, Gs1, Gs2, Gs3);

controlSystemDesigner(Gs, C1s)
```



The corresponding controlSystremDesigner session file is **`multimodel_solution_2025_06_06_FA_gr_A_ControlSystemDesignerSession.mat`**